(How) Will Stablecoins Disrupt Banking?

Bank competition, financial stability and the changing uses of money

Lin William Cong

CCDS & NBS, Nanyang Technological University
ABFER, CEPR, IC3, & NBER

3rd Crafting the Cryptoeconomy Conference
Columbia University · October 29, 2026

Stablecoins and banking: two propositions

Cong (2025), Deposit Dynamics, Financial Stability, and Regulatory Design

- **1. Bank competition** Stablecoins need not reduce deposits or lending. Bank repricing and the GENIUS framework can limit yield-driven disintermediation.
- **2. Financial stability** The principal risk mechanisms are familiar. Reserve, redemption and operational design determine how they transmit.
- **Scope** Fully reserved, fiat-backed payment stablecoins under a prudential framework.

White-paper well-debated: Coinbase, Paradigm, PayPal and Stripe VS. Banking Association

Deposit competition changes banks' behavior

- **Initial friction** Deposit-market power and customer switching costs allow banks to pay less than the competitive return.
- **Bank response** A credible outside option can induce higher deposit rates and better services.

- **Intermediation** Under the competition mechanism, deposits and lending can expand as franchise margins narrow.
- **Model provenance** The white paper adapts the CBDC benchmark of Chiu et al. (2023).

Banks’ deposit rates influence stablecoin demand

Ahmed & Aldasoro (2026) · U.S. weekly data, January 2019–June 2025

Change in deposit rate

+1 bp

An increase in the rate paid by banks

Stablecoin growth over ten weeks

–2.8 pp

Aggregate OLS estimate

USDC –4.8 pp

USDT –1.9 pp

Local projections controlling for market rates, Bitcoin, dollar and VIX; N = 324 at 10 weeks.

Deposit repricing influences stablecoin demand; the credit response depends on banks’ equilibrium adjustments.

Ahmed & Aldasoro (2026), BIS Working Paper 1382, Table OA.A.1 and §5.

The competition benchmark has three regions

Qualitative mechanism in the white paper's Figure 2

Outside-option remuneration	Banking response in the benchmark
Low	Little change in deposit pricing or intermediation
Moderate	Bank repricing can raise deposits and lending
High	Direct substitution can dominate the competition effect

Competition can expand deposits and lending before substitution dominates.

GENIUS constrains yield-driven deposit competition

Provision / mechanism	Implication for the white paper's argument
Issuer interest/yield restriction	Limits direct interest competition with bank deposits
At least 1:1 eligible liquid reserves	Constrains reserve risk and reserve-funded returns
Redemption and disclosure framework	Supports confidence in the payment claim
Remaining adoption margins	Convenience, distribution and rewards need separate analysis

GENIUS narrows direct yield competition while leaving convenience and distribution as adoption margins.

Recalibrating bank competition to the post-2020 rate environment

2024 observed rates

Return on reserves	**5.21%**
Average deposit rate	**1.66%**
Difference	**3.55 pp**

Bank deposit rates rose much less than returns on reserves.

Updated inputs in Cong (2025)

Production curvature	**0.697**
Reserve requirement	**0%**
Return on reserves (average)	**3.9%**
Inflation / money growth	**4.2%**
Cash / digital / mixed (%)	**3.27 / 31.9 / 64.83**

Maintained structure: 26 banks, trading probability 0.56, and the remaining baseline parameters.

Competition can expand banking intermediation

Illustrative 2025 calibration, conditional on the maintained model and institutional regime

Remuneration region	Deposit and lending response reported in the paper
Low	Approximately unchanged
About 4–6%	Competition supports deposits and lending
Above about 6%	Substitution can dominate and reduce intermediation

The 4–6% region supports intermediation under the paper's stated calibration.

Banks adjust along different margins

Bank characteristic	Relevant adjustment
Deposit pricing and customer sensitivity	Reprice balances and change service quality
Funding and portfolio flexibility	Replace funding or adjust liquid assets
Payment and custody capability	Supply complementary services to issuers and users
Borrower relationships	Preserve information and manage migration costs

Bank funding responses vary; deposit reallocation and aggregate credit can follow different paths.

The principal financial risk mechanisms are familiar

Stablecoin concern	Familiar financial mechanism
Redemption run or depeg	Runs and coordination problems in redeemable claims
Reserve or custodian impairment	Asset, counterparty and custody exposure
Redemption faster than asset realization	Liquidity and settlement mismatch
Concentrated reserve sales	Market-capacity limits and fire-sale spillovers
Failure to transfer or reconcile claims	Operational risk in payment infrastructure

Established prudential principles provide a starting point for design.

Payment information, liquidity and monetary trust

Contribution	Economic margin	Question for banking
Parlour, Rajan & Zhu	Payment information	Which projects obtain finance?
Lee & Tou	Issuer-related liquidity demand	How does incoming funding change bank portfolios?
Harris & Rogoff	Institutions behind private money	What sustains trusted par redemption?

The mechanisms connect bank funding, information in payment flows and trust in private money.

Common shocks can propagate through onchain links

- **Timing** Continuous token transfers can meet constraints in fiat redemption and settlement.
- **Concentration** Large flows may depend on shared custodians, dealers and operating providers.
- **Scale** Greater normal-time participation can coexist with larger common stress exposures.
- **Research question** How do connected issuers, banks, protocols and markets amplify a common shock?

The research task is to trace shocks through connected issuers, banks and protocols.

Safeguards should address the mechanism that creates the loss

Economic problem	Relevant safeguard
Reserve loss or opacity	Eligible assets, valuation and credible reporting
Custody or legal uncertainty	Segregation, controls and clear priority
Redemption liquidity	Cash access, liquid reserves and contingency arrangements
Operating failure	Continuity, reconciliation and settlement capacity
Concentrated market stress	Counterparty and liquidity-concentration management

Prudent integration can improve banking and payments

Potential gain	Condition for realizing it
Better payment services	Reliable access, redemption and interoperability
Less settlement exposure	Enforceable completion and sufficient liquidity
Bank service opportunities	Fees cover custody, operating and liquidity costs
More efficient intermediation	Competition improves terms without unmanaged risk

Payment innovation and banking can reinforce one another when the underlying exposures are managed.

Five Facts About Stablecoins

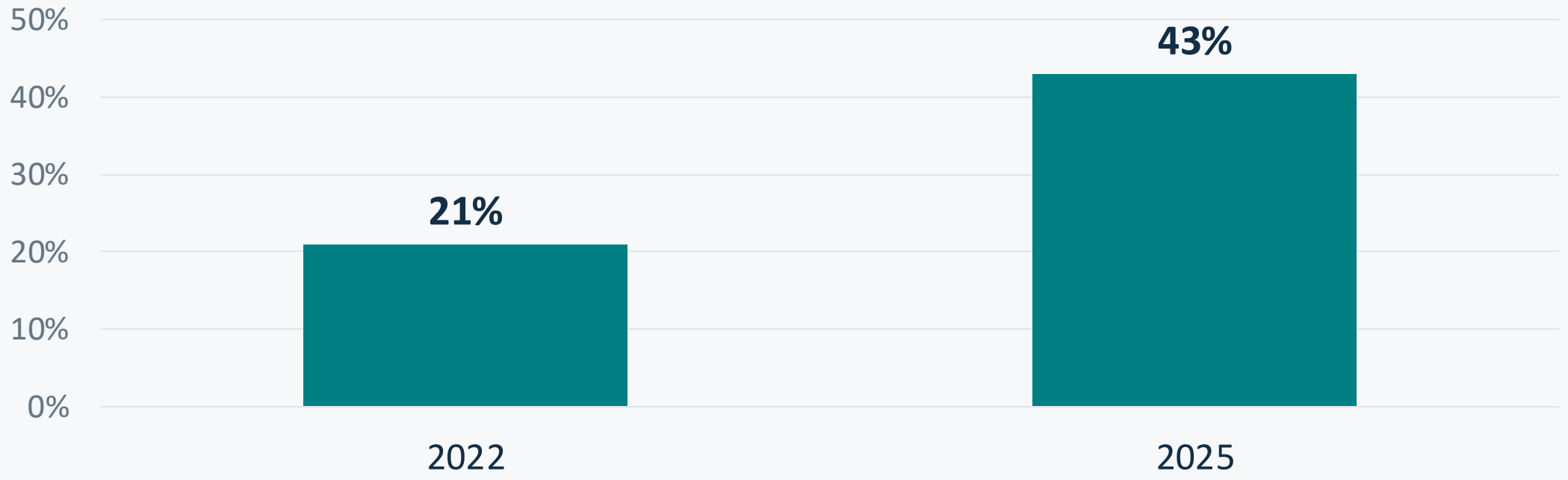
Cong, Liao, Prasad & Zhang (2026): a separate empirical paper

Fact	Data / method	Banking margin
1. Transaction currency	Transfers and size distributions	Payment services
2. Global adoption	Timing-based location probabilities	Cross-border funding
3. Cash management	Wallet moments and a structural benchmark	Demand for balances
4. Onchain lending	Protocol and platform borrowing	Credit infrastructure
5. Programmable payments	Contract and facilitator classifications	Automated settlement

Cong et al. (2026), Sections 3–4 and Appendices A–D. Gordon Liao is affiliated with Circle

Fact 1: Stablecoins are increasingly a transaction currency

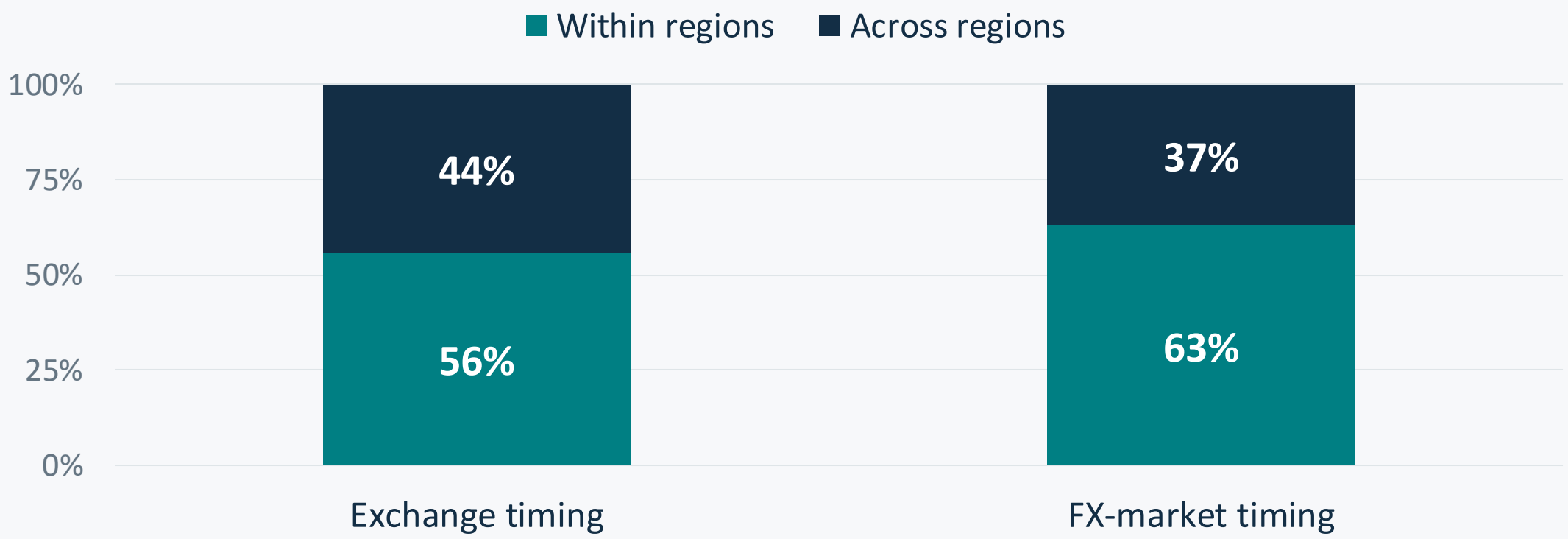
Transfers below $1 / filtered transfer count



USDC and USDT transfers on covered chains (excluding Solana); exchange and contract addresses filtered.

Transfers below $1 rose from 21% to 43% of the filtered count between 2022 and 2025.

Fact 2: Stablecoin use spans the major world regions



2025 Ethereum wallet-to-wallet value; Bayesian attribution based on transaction timing.

In 2025, 37–44% of selected wallet-to-wallet value crossed the three broad regions.

Fact 3: Stablecoin wallets are managed like cash

Alvarez–Lippi cash management: replenishment with fixed costs and free adjustment opportunities

$dm/dt = -c$ $\alpha = b/c$ $mpre/M = p/n$

Model and estimation

- Holding cost: foregone return r. Paid adjustment cost: b.
- Restore to a return level at a free opportunity or at zero.
- Fit 3 wallet moments to 2 parameters: p and α.

Banking implication

Payment convenience affects desired working balances, independently of nominal remuneration.

Wallet adjustment costs and precautionary balances vary across user types.

Fact 3: Typical wallets hold ten days of outflow

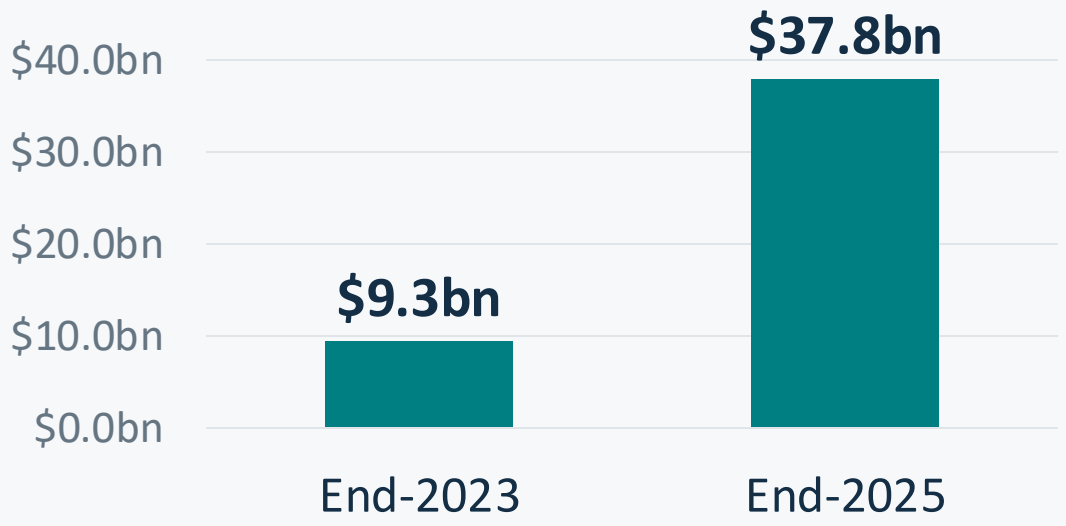
Selected Ethereum wallets; daily balances over the 183 days ending December 31, 2025

Empirical moment	Geometric mean	Interpretation
M/c: holdings / daily outflow	10.00 days	Working-balance intensity
n: replenishments	10.1 per year	Frequency of balance adjustment
mpre/M: pre-replenishment balance	0.25	Precautionary holdings

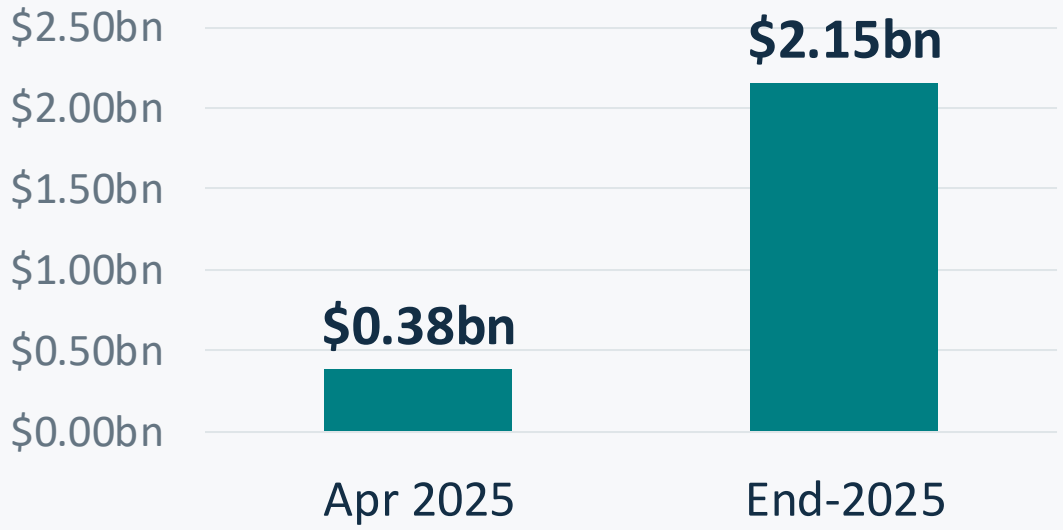
Fitted access costs: 6.2% of daily outflow overall; 17.5% among inventory wallets.

Fact 4: Onchain credit markets are expanding

Protocol borrowing ($bn)



Selected platforms ($bn)



Sources: DeFiLlama protocol borrowing (all assets); Maple, Goldfinch and TrueFi loans (selected platforms).

Secured protocol borrowing quadrupled in two years; institutional onchain lending is also growing.

Fact 5: Stablecoins enable programmable payments

USDC and USDT on Ethereum, 2025

Classification	Transaction-hash share	Gross transfer-value share
Any contract involvement	35%	69%
Contract sender only	28%	60%

Composition Of labeled contract-involved value: lending 33%; market making/OTC 25%; custody 22%; payments/commerce 0.1%.

Early x402 activity demonstrates machine payment feasibility; sustainable commercial demand remains to be measured.

Most contract-mediated value reflects financial automation; agentic commerce is an emerging application.

What Five Facts implies for banking

Observed margin	Banking object still to identify
Payment activity	Useful purchases, all-in costs and customer relationships
Wallet geography	Displaced asset and full funding chain
Balance management	Stable funding versus transitory settlement balances
Onchain credit	Borrower substitution, maturity, collateral and losses
Programmable transfers	Delegation, durable use and contingent liquidity

Next research questions: network-wide stress transmission and the quality of delegated agent decisions.

Public money responds to private payment innovation

Cong research	Relevant contribution	Boundary
Bai, Cong, Luo & Xie (2025)	Early e-CNY adoption and promotion evidence	Pilot associations, not a 2026 credit effect
Cong & Mayer (2025)	Strategic digitization under monetary competition	A conditional model, not a lending forecast

The counterfactual includes endogenous improvements to public and bank payment systems.

e-CNY in 2026: remuneration with bank-deposit treatment

Commercial-bank wallet arrangements effective January 1, 2026

Design dimension	Documented arrangement
Liability	Commercial-bank deposit treatment
Remuneration	Posted demand-deposit rate for eligible identified wallets
Prudential treatment	Normal reserves and deposit-insurance framework
Inference for banking	Adoption and bank participation can change jointly

Interest rises from zero. The sources do not establish a yield premium over ordinary deposits.

PBOC policy reported by gov.cn, Dec. 29, 2025. CCB announcement, Dec. 31, 2025; Type IV excluded

Agentic payments and banks' liquidity services

- **Cash management** Automated adjustment may reduce idle payment balances and increase replenishment frequency.
- **Liquidity timing** Common execution rules may concentrate payments and redemptions at the same time.
- **Banking services** Credit lines, cash management and settlement access can remain valuable even when payment execution moves onchain.
- **Evaluation question** Which delegated agent policies remain robust to liquidity shocks and strategic responses?

Delegated payments: credit and enforceable claims

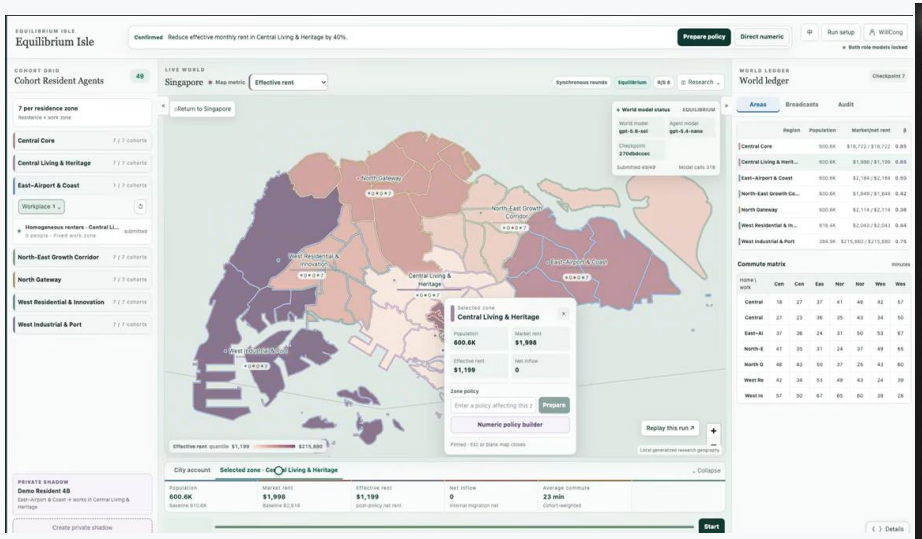
Agent or tokenized-asset need	Intermediation problem
Spending authority and budget	Identify the principal and enforce limits
Contingent credit	Underwrite repayment and fund drawdowns
Collateral and settlement	Verify ownership, valuation and legal priority
Failure or dispute	Allocate losses and provide enforceable recourse

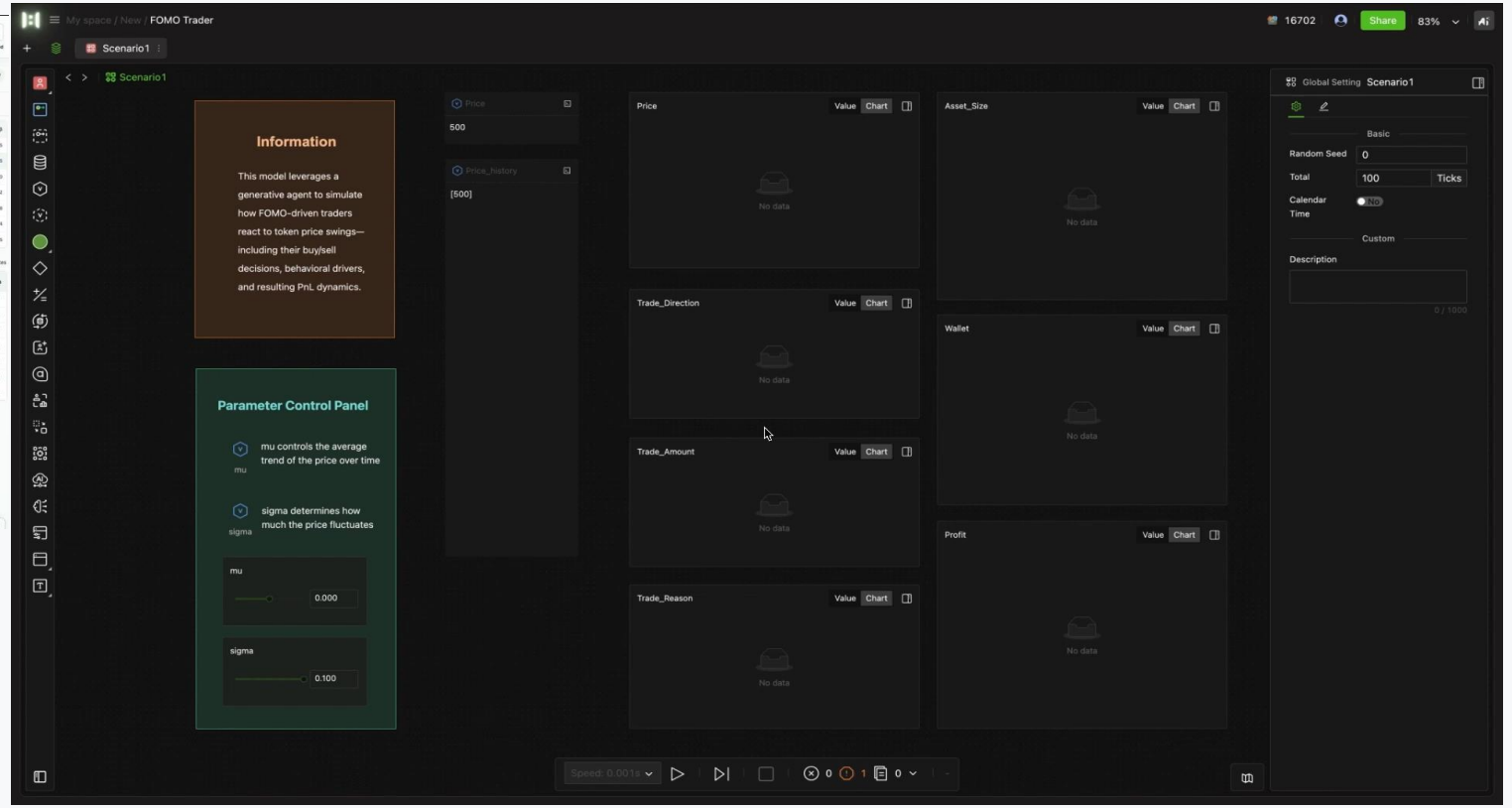
Delegation links payment authority to credit, ownership and enforceable claims.

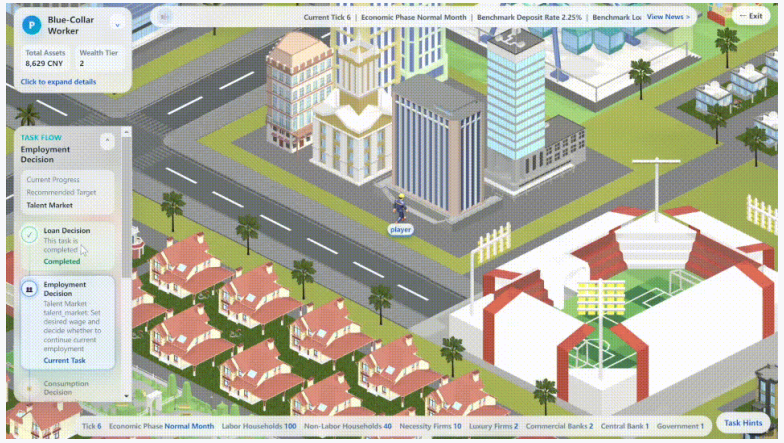
Cong, Mayer & Rabetti (2026), Tokenizing Real-World Assets. AP2 is an architecture example

Economic World Models Initiative

Economic World Models (EWM) | an integrated laboratory for mechanisms and stress propagation







Global Metacase Initiatives

Global MetaCase Initiative | economic decision quality, incentives and institutional constraints

MetaCase: state and data → objective and authority → decision → outcome

Global Metacase Challenge

Scientific test: does individual competence survive interaction with other autonomous agents?

